Final Documentation: Deviations from SRS/SAD

Student Record System v1.0 · Team 12 · Software Engineering mini project GitHub issue #39 · Sprint 2 (26–30 Oct 2026) · Status: DRAFT, 1 Nov 2026

1. Purpose and method

This document lists 12 places where the delivered system (final `main` branch) differs from the SRS v1.0 (11-09-2026) or the SAD v1.1 (08-10-2026), with the clause each one breaks, the evidence, and the proposed handling. Findings come from comparing both documents with the source, headers, tests, README and the Software Test Plan (STP) with its RTM.

Severity: High = a High-priority SRS requirement is not met. Medium = a Medium-priority SRS requirement or a stated SAD design rule is not met. Low = layout, naming or documentation only. Each entry gives the most likely reason, drawn from the code, README and the two documents.

2. Summary

ID	Deviation	Breaks	Severity	Handling
D-01	Student update and delete are not in the console menu	SRS STU-F-011 (High), STU-F-012; SAD 4.5	High	Fix
D-02	Deleting a student asks for no confirmation	SRS STU-F-012 AC; SAD 4.4	Medium	Fix
D-03	Data file permissions are not enforced on Windows	SRS STU-SR-005, AC-SEC-005	Medium	Accept and document
D-04	grades.csv is rewritten in place, not safely	SRS STU-NF-002; SAD 3.4, 3.7	Medium	Fix
D-05	No usability walkthrough on record	SRS STU-NF-004, TC-UX-01	Medium	Run or document
D-06	No unlock or account management beyond create login	SAD 4.3 reset_lockout(); SRS 7.1	Medium	Accept, update SAD
D-07	Student "Update Own Profile Details" not implemented	SRS 7.1; SAD 4.7	Low	Update SRS
D-08	Student and attendance file layouts differ from SAD 4.6	SAD 3.6, 4.6	Low	Update SAD
D-09	Grades, attendance and audit bypass StorageService	SAD 3.3, 3.4	Low	Update SAD

ID	Deviation	Breaks	Severity	Handling
D-10	Grade module does not check that the student exists	SAD Figure 3	Low	Fix or update SAD
D-11	Audit event and function names differ from the SAD	SAD 3.5, 4.3	Low	Update SAD
D-12	Test plan and RTM are out of date	STP 13; SRS 8	Low	Update

3. Deviations

D-01 Student update and delete are not in the console menu

- **Required:** STU-F-011 (High) and STU-F-012; SAD 4.5 lists Add, update, delete, search and view under Student Records.
- **Delivered:** `update_student()` and `delete_student()` exist, are admin-only, audited and unit-tested (TC-Rec-02, TC-Rec-03). The admin menu has only *Add student record* and *View student record*. The README already says update and delete are "in the service layer".
- **Impact:** an administrator cannot correct or remove a student in the running application, so these acceptance criteria cannot be shown at system level.
- **Handling:** fix. Two menu entries calling the existing functions close it.

Reason: sprint effort went to the demo workflow (attendance, grades, reports, student login). The service functions were built and tested first, and the menu entries were left for later.

D-02 Deleting a student asks for no confirmation

- **Required:** STU-F-012 acceptance criterion: removed only after explicit confirmation. SAD 4.4: record deletion requires explicit confirmation and an audit entry.
- **Delivered:** `delete_student()` deletes immediately and writes the audit entry. No confirmation prompt exists anywhere in the code. The RTM row for TC-Rec-03 says confirmation logic was verified, which the code does not support.
- **Handling:** fix together with D-01 (a y/N prompt in the menu), then correct the RTM row.

Reason: follows from D-01. Delete was built as a service-layer function, and the confirmation belongs in the menu that was never added.

D-03 Data file permissions are not enforced on Windows

- **Required:** STU-SR-005 and AC-SEC-005: no world-readable or world-writable data file.
- **Delivered:** Linux and macOS files get mode 0600. On Windows the program only clears the read-only flag and relies on the folder's inherited NTFS permissions. The comment in `fileperm.h` says so.
- **Impact:** the criterion cannot be shown on Windows, although the SRS lists Windows as a target platform.

- **Handling:** accept and document as a platform limitation; setting an explicit ACL would be future work.

Reason: Windows has no Unix permission bits, and the application uses only the portable C library. Setting NTFS ACLs would need Windows-specific code that the Linux build could not share.

D-04 grades.csv is rewritten in place

- **Required:** STU-NF-002: no loss or corruption after an abnormal exit. SAD 3.4 and 3.7: write to a temporary file, flush, then replace.
- **Delivered:** the student, user and attendance files follow this. `grade.c` opens `grades.csv` for writing, which truncates it, then writes all rows. A crash in between can lose every grade. TC-Rel-01 exercised the student file only.
- **Handling:** fix by reusing the temp-file-and-rename approach from `attendance.c`, and add a grade case to the reliability test.

Reason: the grade module was written as a self-contained module with its own file handling. The safe-write pattern was applied to the other files but not retrofitted here.

D-05 No usability walkthrough on record

- **Required:** STU-NF-004: three test users, at least 90% task success (TC-UX-01).
- **Delivered:** no TC-UX-01 test, results or notes were found in the repository, and the RTM has no row for it.
- **Handling:** run a short walkthrough with three people and record it, or list it as not performed. Owner to confirm that nothing was done outside the repo.

Reason: project time went to implementation and automated tests. A walkthrough needs three outside users and was not scheduled.

D-06 No unlock or account management beyond creating a login

- **Required:** SAD 4.3 lists `reset_lockout()` as planned. SRS 7.1 shows *Manage User Accounts* for the administrator, and SAD 3.8 maps it to add user and default administrator.
- **Delivered:** `auth_add_user()` and the default admin exist. Nothing unlocks an account, resets a password or removes a user. A locked account, including the only admin, stays locked until `users.dat` is edited by hand.
- **Impact:** STU-F-003 itself is met (lock and log). The gap is the planned recovery path.
- **Handling:** accept and mark `reset_lockout()` as not delivered in the SAD, or add an admin unlock option.

Reason: the SRS only requires locking and logging (STU-F-003), so recovery was treated as lower priority than the core requirements.

D-07 Student "Update Own Profile Details" is not implemented

- **Required:** the SRS 7.1 student use-case diagram includes it, but no functional requirement does. SAD 4.7 already flags this for reconciliation.
- **Delivered:** students are read-only (STU-F-004), as the SAD proposed.

- **Handling:** remove the use case from the SRS diagram.

Reason: the SRS requirement list never included profile editing, and SAD 4.7 already proposed removing the use case. Version 1 keeps students read-only.

D-08 Student and attendance file layouts differ from SAD 4.6

- **Required:** SAD 3.6 and 4.6: every data store is a delimited text file, one record per line. Attendance status is "present or absent".
- **Delivered:** `students.dat` holds fixed-size binary structs. `attendance.csv` stores status as 0 (present) or 1 (absent). The users file, grades file and audit log match the SAD.
- **Impact:** the student file cannot be read or edited as text, and its layout depends on compiler struct padding, which matters for STU-NF-005. SAD 4.7 left these layouts "to be confirmed", so updating the SAD is enough.
- **Handling:** update SAD 4.6 to the delivered formats; note the portability caveat.

Reason: SAD 4.6 marked these layouts as proposed and left them to the owning modules. A fixed-size binary struct made updating and deleting a student record simple, and a numeric status kept attendance rows compact.

D-09 Grades, attendance and audit bypass StorageService

- **Required:** SAD 3.3 and 3.4: all persistence goes through StorageService. SAD 4.7: move the user store and auth logging onto StorageService and AuditService.
- **Delivered:** the user store and auth logging were moved (issue #33). The student file uses StorageService. `grade.c`, `attendance.c` and `audit.c` do their own file I/O.
- **Impact:** safe-write behaviour is not uniform, which is how D-04 arose.
- **Handling:** update the SAD component description, or move the writes behind StorageService.

Reason: modules were built in parallel by different members, each owning its own file handling. StorageService was integrated for users and students first.

D-10 Grade module does not check that the student exists

- **Required:** SAD Figure 3: the grade module asks the record module whether the student exists and returns STUDENT_NOT_FOUND.
- **Delivered:** `record_grade()` accepts any valid roll number. The existence check is in the console function `do_enter_grade()`, and the grade module has no such result code.
- **Impact:** any other caller of the grade module can store grades for a non-existent student.
- **Handling:** move the check into the grade module, or update Figure 3.

Reason: the console menu already checks the student before calling the grade module, so the check ended up in the UI during integration.

D-11 Audit event and function names differ from the SAD

- **Audit events:** the code logs `LOGIN_BLOCKED_ACCOUNT_LOCKED` (SAD: LOGIN_BLOCKED), `RECORD_GRADE` and `UPDATE_GRADE` (SAD: GRADE_SAVE), and an extra

`LOGIN_FAILED_UNKNOWN_USER`. Student adds, updates and deletes are logged under module APP, while denied attempts are logged under STUDENT.

- **Functions:** `get_student` (SAD: find_student), `search_students` (search_student), `list_students` and `grade_list_for` are not in the SAD interface table.
- **Handling:** update the SAD tables and the sequence diagrams.

Reason: modules were built in parallel and named independently, and the SAD was written before the final names were fixed.

D-12 Test plan and RTM are out of date

- Rows for STU-SR-003 and the grouped security cases still say "awaiting integration" and mention a null-session defect. `record_service.c` now rejects null and student sessions and audits the attempt, so these should be re-run.
- The RTM has no rows for STU-F-021 to 023, 031 to 033, 041 to 042, STU-NF-004 or STU-NF-005, although tests exist for most (see `docs/test_cases_member4.md`).
- Requirement IDs appear as `STU-F001` in the SRS tables, `STU-F-001` in the SRS text and STP, and `STU-F001` in code. The SAD header says version 1.1 while its revision history lists only 1.0.
- The SAD says the RTM was at v1.2; that version is not in the repository. The STP is still marked Draft, with the approvals table unsigned.

Reason: the STP and RTM were last revised in September, and later fixes and tests landed after that.

4. Choices that are within the SRS/SAD

These were checked and are not deviations, but a reader may wonder about them.

- **Password masking differs by platform.** Windows shows `*` and caps entry at 64 characters (TC-Auth-02M). Linux and macOS show nothing. STU-F-002 only requires that characters are not echoed, so both comply.
- **SHA-256 without salt, default admin.** SRS STU-SR-001 asks for a standard one-way hash, and SAD 3.4 names a default administrator. There is no change-password feature and the default is `admin` / `admin123`; state this as a known limitation.
- **Storage is flat files.** SRS 2.4 and SAD 3.6 allow files or SQLite and name CSV as the v1 baseline.
- **Capacity limits.** The 100-user, 31-character username, 64-character password and 15-character roll number limits match SAD 4.6. The 1,000-grade limit is not stated in either document.
- **Grade scheme.** The A to F bands (90, 80, 70, 60, 50) match the SAD 4.6 proposal exactly. SAD 4.7 still lists confirmation with the course instructor as open.
- **Audit log has no viewer.** STU-SR-003 requires entries, not a screen; the log is read from `audit.log`.
- **Extras beyond the SRS:** partial, case-insensitive name search; attendance below 75% flagged LOW; CSV export that neutralises spreadsheet formulas.

5. Delivered as specified

Login, hashing, three-failure lockout that persists across restarts, and role-based access (STU-F-001 to 004). Add, view and duplicate rejection for students (STU-F-010, 013). Marks entry, update and letter

grades (STU-F-020 to 022). Attendance marking, correction and percentage (STU-F-030 to 032). Search, grade sheet, attendance summary and CSV export (STU-F-040 to 042). Response time (p90 0.003 s against 2 s), crash-safe writes for student, user and attendance files, input validation, audit logging, and Windows and Linux builds through CI.